

OPTIMIZING PARKING DISTRIBUTION IN HONG KONG

Background

Hong Kong's vehicle parking system is riddled with problems. The most long-standing issue is the shortage of parking spaces.

The problem of insufficient parking spaces in Hong Kong is expected to worsen with the recently raised "Southbound Travel Scheme" which will be implemented in less than a month, putting even greater pressure on traffic systems while increasing the number of vehicles in Hong Kong.

Furthermore, mainland drivers may not be familiar with Hong Kong's parking system. Due to the different driving systems between two regions – The absence of parking meters – mainland drivers will also need to adapt to the parking system in HK as soon as possible.

Finally, with the increasing popularity of electric vehicles (EV) in mainland China and Hong Kong, there is no unified information platform that provides information about both parking meters and EV charge spots.

Approach and Methods

In response to the above situation, our plan is to create a data platform to integrate parking information and electric vehicle charge spot information in Hong Kong. It will also provide a user-friendly UI to facilitate drivers and reduce the load on the parking system.

Therefore, we developed 4 major functions with our database and webUI:

First, it provides the locations, operating hours and types of all paid parking meters across Hong Kong with visualised display - with both list and map.

Second, it displays the locations and basic information with a list or map of all public electric vehicle charging stations.

Third, it encourages users to report actual electric vehicle charging prices, allowing the public to access the latest fee information, as no publicly available data currently exists.

Fourth, it presents the density distribution of parking spaces and electric vehicle charging stations in each district through maps and charts.

To achieve the functions, we found a data set that contains Parking Meters

information and a set of data that contains EV Charge Spots information. Then we integrate this data and transform it as an SQL database for use by the webUI.

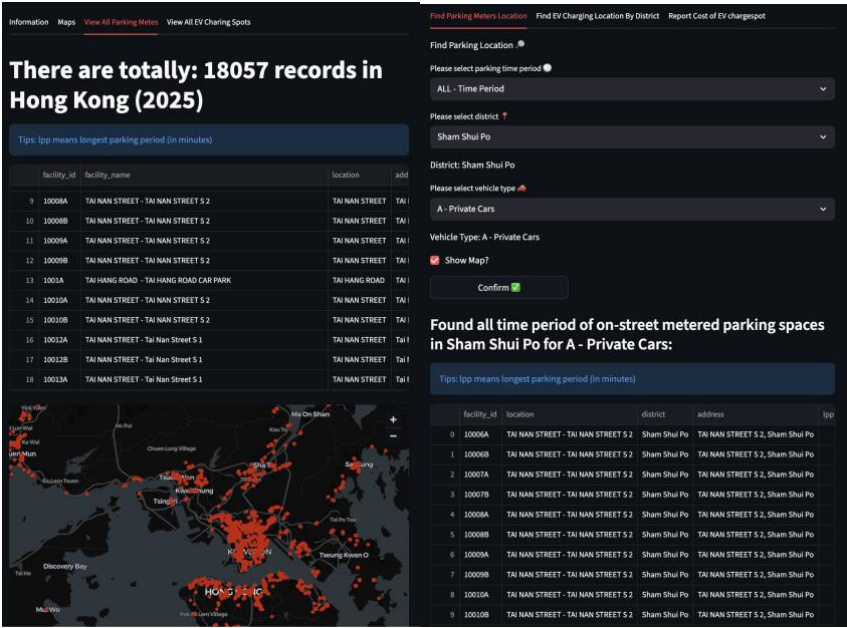
Platform Design

Our platform offers mainly 3 query functions:

1. Comprehensive Parking Meters Locator:

This is the main application of our platform. It provides a user-interface for users to search for various types of metered parking locations across Hong Kong, including service hours and availability — supported by both data tables and map visualization.

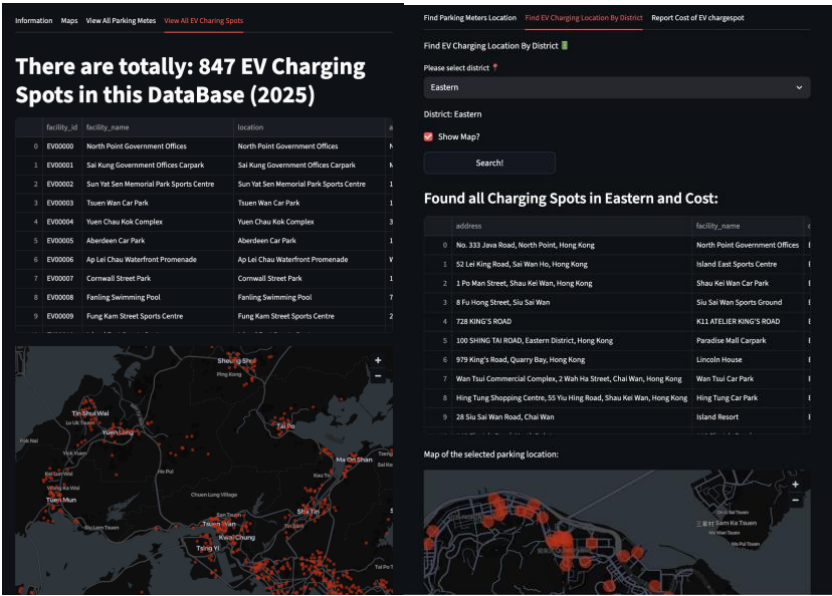
In addition, the UI displays the total number of parking meters and EV charging stations in our database. A heatmap is also provided to show the distribution density of both facilities. This allows users to clearly see the distribution and monitor the development of Hong Kong's parking system.



2. EV Charging Meter Identification:

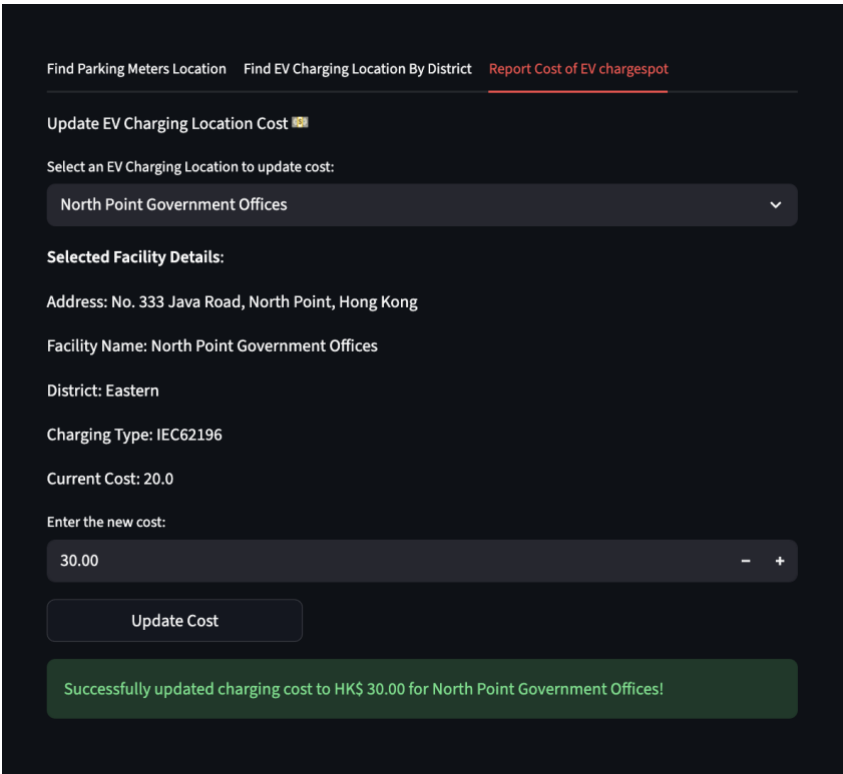
The update will be immediately synchronized to EV Charging Meter Identification for

review, benefiting both yourself and others.



3. User-driven Feedback System:

A built-in feedback mechanism allows users to report real-time updates on EV charging spot cost. Updates will be synchronized immediately for your review, benefiting both yourself and others.



Database Design

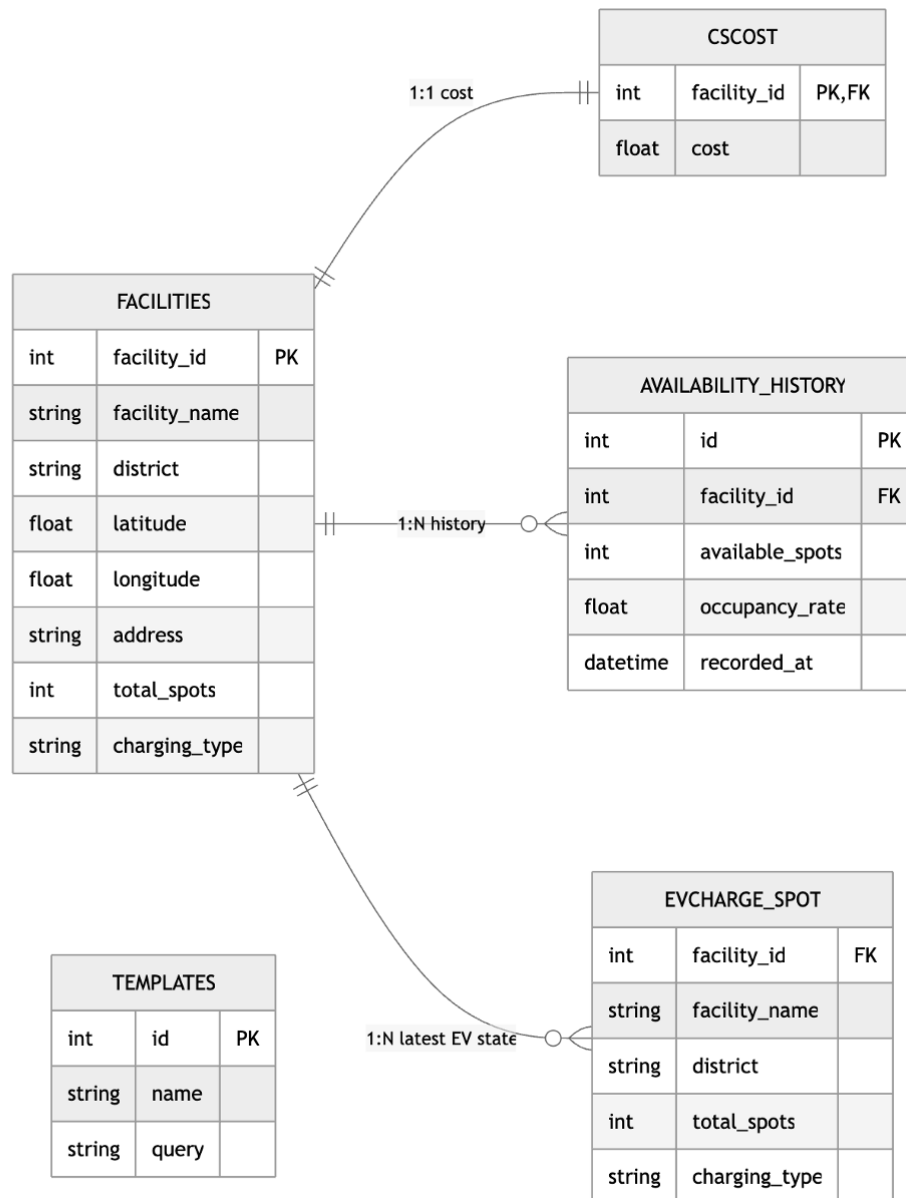
To build the core of the automated content management system, we designed and implemented a robust and normalized SQLite database. The design ensures data

integrity and supports efficient report generation queries.

ER Model

Our system is built on an ER model that establishes relationships between different data entities. This model centers around the Facilities table, which serves as the primary entity for all parking and charging locations.

The diagram below shows the relationships between core tables:



Facilities Table: Uses facility_id as the primary key.

EVCHARGE_SPOT Table: Uses facility_id as a foreign key, establishing a one-to-many relationship with the Facilities table.

Other Tables:

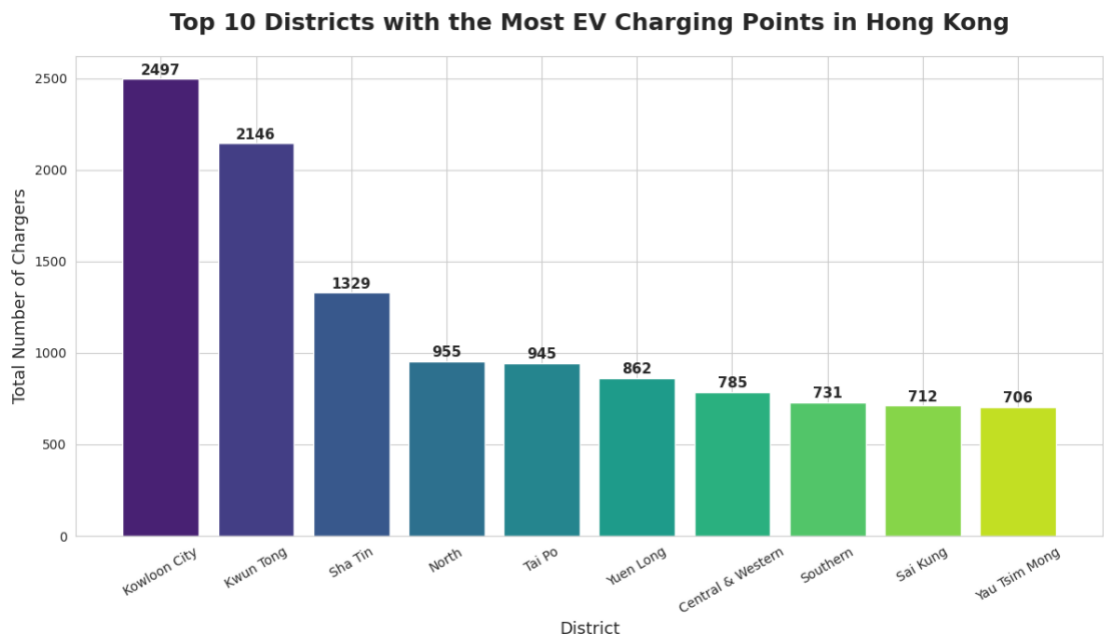
Availability_History Table: References the evcharge_spot table, establishing a one-to-

many relationship.

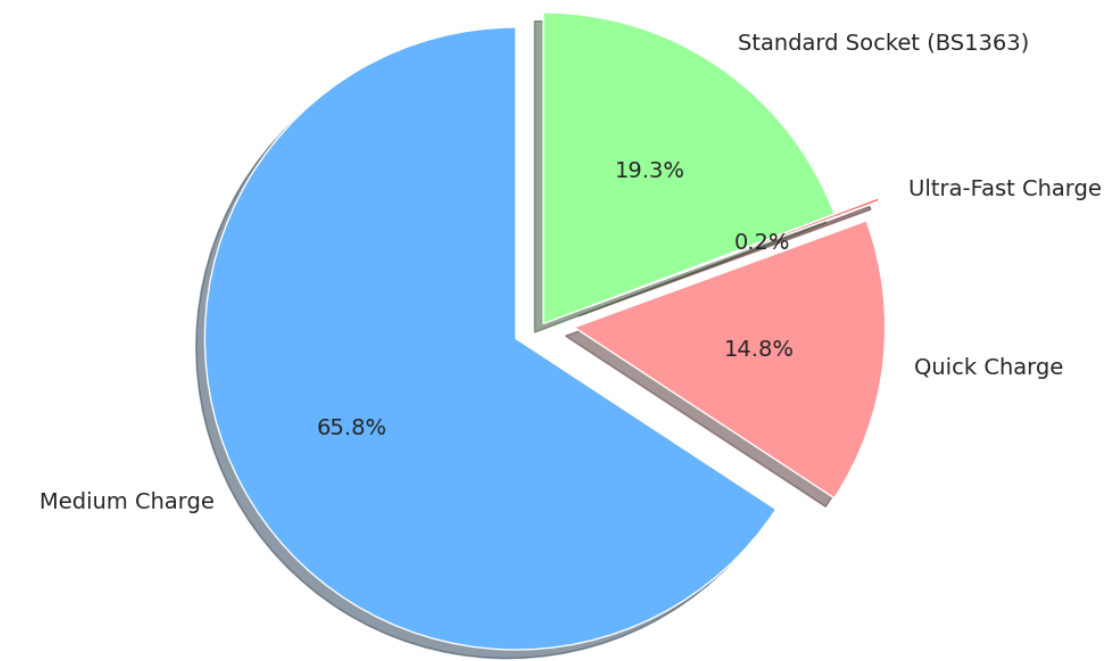
Facilities_CSCOST Table: Establishes a one-to-many relationship using facility_id.

We design the database this way to maintain relationships between tables to support structured and efficient data management.

Findings



Distribution of EV Charger Types in Hong Kong (Latest 2025 Data)



After compiling and processing the data, we obtained following key figures:

The total metered parking space in Hong Kong is 18,057, while the total public charging points is 16,342. Among the 18 districts in Hong Kong, Kowloon City has the highest number of charging points of 2,497.

Limitations

There are some limitations in our project:

1. Cost data is not yet provided due to the lack of open-source data.
2. Some districts show no EV chargers, which is inconsistent with the EV charger dataset. This reflects gaps in the official 2025 dataset provided on the government website.

Future development

In regard of our current project, we plan to modify our product in the following ways:

1. Partnering with proprietors to get the live occupancy data of park meters and charge spots.
2. Partnering with proprietors to get comprehensive information about the car park, so our platform can provide car park information not only parking meters.
3. Build time-series forecasting models to predict peak parking periods, for drivers' reference.

Reference & Data Source

Reference:

1. Registration and Licensing of Vehicles and Drivers

https://www.td.gov.hk/mini_site/atd/2025/en/section3-2.html

2. Total number of parking spaces in the 18 districts by vehicle type over the past 3 years

https://gia.info.gov.hk/general/202506/11/P2025061100254_497951_1_17496322816

[41.pdf](#)

Dataset:

(Core Dataset Source)

1. Distribution of Metered Parking Spaces in Hong Kong Districts

Provider: Transport Department, Hong Kong Special Administrative Region (HKSAR)

Acquisition Platform: Hong Kong Government Open Data Platform -

https://data.gov.hk/tc-data/dataset/hk-td-tis_4-distribution-of-metered-parking-spaces

(Supporting Platform)

2. Metered Parking Spaces in Hong Kong

https://opendata.esrichina.hk/datasets/3d89ec348c684473a86ca6c8083af924_0/about

3. Electric Vehicle Chargers for Public Access

<https://data.gov.hk/en-data/dataset/hk-epd-evcpateam-evc-1/resource/2eb2abc9-4fc1-4344-b68e-df2e2c69fc00>

Data Format: Supports two formats (CSV and XLSX) that can be directly imported into the database.